PHENOL **ICSC: 0070 (April 2017)**
Carbolic acid
Phenic acid
Hydroxybenzene

CAS #: 108-95-2
UN #: 1671
EC Number: 203-632-7

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Combustible. Above 79°C explosive vapour/air mixtures may be formed.	NO open flames. NO contact with strong oxidizing agents. Above 79°C use a closed system and ventilation.	Use water spray, alcohol-resistant foam, powder, carbon dioxide.

AVOID ALL CONTACT! FIRST AID: USE PERSONAL PROTECTION. IN ALL CASES CONSULT A DOCTOR!

	SYMPTOMS	PREVENTION	FIRST AID
Inhalation	Sore throat. Burning sensation. Cough. Dizziness. Headache. Shortness of breath. Laboured breathing. Unconsciousness. Symptoms may be delayed. See Notes.	Avoid inhalation of dust and mist. Use ventilation, local exhaust or breathing protection.	Fresh air, rest. Half-upright position. Refer for medical attention.
Skin	MAY BE ABSORBED! Serious skin burns. Numbness. Convulsions. Collapse. Unconsciousness.	Protective gloves. Protective clothing.	Wear protective gloves when administering first aid. Remove contaminated clothes. Rinse skin with plenty of water or shower. To remove substance use polyethylene glycol 300 or vegetable oil. Refer immediately for medical attention .
Eyes	Pain. Redness. Loss of vision. Severe burns.	Wear face shield or eye protection in combination with breathing protection.	Rinse with plenty of water for several minutes (remove contact lenses if easily possible). Refer immediately for medical attention.
Ingestion	Sore throat. Burns in mouth and throat. Convulsions. Abdominal pain. Diarrhoea. Shock or collapse.	Do not eat, drink, or smoke during work. Wash hands before eating.	Rinse mouth. Give one or two glasses of water to drink. Do NOT induce vomiting. Refer immediately for medical attention.

SPILLAGE DISPOSAL

Personal protection: chemical protection suit including self-contained breathing apparatus. Do NOT let this chemical enter the environment. Sweep spilled substance into covered sealable containers. If appropriate, moisten first to prevent dusting. Carefully collect remainder. Then store and dispose of according to local regulations.

STORAGE

Provision to contain effluent from fire extinguishing. Separated from strong oxidants and food and feedstuffs. Dry. Well closed. Store only in original container. Keep in a well-ventilated room. Store in an area without drain or sewer access.

PACKAGING

Do not transport with food and feedstuffs.

CLASSIFICATION & LABELLING

According to UN GHS Criteria

DANGER

Toxic if swallowed or in contact with skin
Causes severe skin burns and eye damage
Suspected of causing genetic defects
Causes damage to central nervous system, the heart and kidneys
Causes damage to organs through prolonged or repeated exposure
May cause respiratory irritation
Toxic to aquatic life

Transportation
UN Classification
UN Hazard Class: 6.1; UN Pack Group: II

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PHYSICAL & CHEMICAL INFORMATION

Physical State; Appearance
COLOURLESS-TO-YELLOW OR LIGHT PINK CRYSTALS WITH CHARACTERISTIC ODOUR.

Physical dangers

Chemical dangers
The solution in water is a weak acid. Reacts with oxidants. This generates fire and explosion hazard.

Formula: C_6H_6O / C_6H_5OH
Molecular mass: 94.1
Boiling point: 182°C
Melting point: 41°C
Density: 1.06 g/cm³
Solubility in water, g/l at 20°C: 84 (moderate)
Vapour pressure, Pa at 20°C: 47
Relative vapour density (air = 1): 3.2
Relative density of the vapour/air-mixture at 20°C (air = 1): 1.0
Flash point: 79°C c.c.
Auto-ignition temperature: 715°C
Explosive limits, vol% in air: 1.3-9.5
Octanol/water partition coefficient as log Pow: 1.46

EXPOSURE & HEALTH EFFECTS

Routes of exposure
Serious local effects by all routes of exposure.

Effects of short-term exposure
The substance and the vapour are corrosive to the eyes, skin and respiratory tract. Corrosive on ingestion. Inhalation of the vapour may cause lung oedema, but only after initial corrosive effects on eyes and/or airways have become manifest. See Notes. The substance may cause effects on the central nervous system, heart and kidneys. This may result in convulsions, coma, cardiac disorders, respiratory failure and collapse. The effects may be delayed. Medical observation is indicated. Exposure could cause death.

Inhalation risk
A harmful concentration of airborne particles can be reached quickly when dispersed, especially if powdered.

Effects of long-term or repeated exposure
The substance may have effects on the liver, kidneys and nervous system.

OCCUPATIONAL EXPOSURE LIMITS

TLV: 5 ppm as TWA; (skin); A4 (not classifiable as a human carcinogen); BEI issued.
MAK: skin absorption (H); carcinogen category: 3; germ cell mutagen group: 3B.
EU-OEL: 8 mg/m^3, 2 ppm as TWA; 16 mg/m^3, 4 ppm as STEL; (skin)

ENVIRONMENT

The substance is toxic to aquatic organisms.

NOTES

Other UN numbers: 2312 (molten); 2821 (solution).
Depending on the degree of exposure, periodic medical examination is suggested.
The symptoms of lung oedema often do not become manifest until a few hours have passed and they are aggravated by physical effort. Rest and medical observation are therefore essential.

ADDITIONAL INFORMATION

EC Classification
Symbol: T, C, Xn; R: 23/24/25-34-48/20/21/22-68; S: (1/2)-24/25-26-28-36/37/39-45